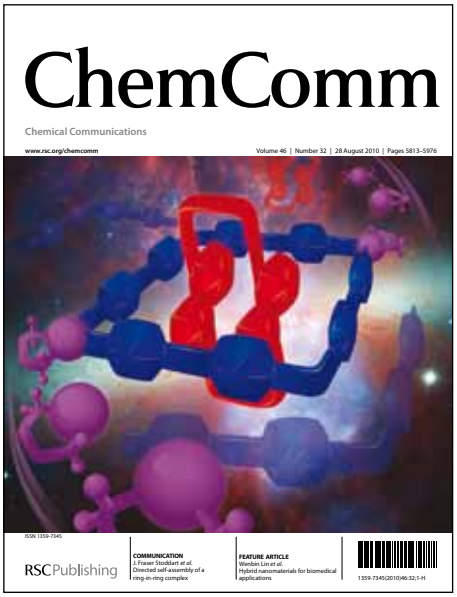
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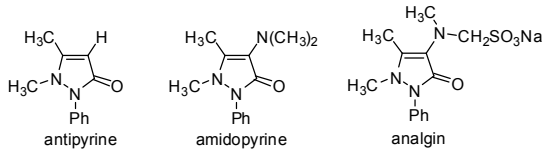
Efficient Synthesis of 1,2-Bis(alkoxycarbonyl)pyrazol-3-ones from 2,3-Allenoic Acids, Azodicarboxylates and PPh_3

Ruizhi Lü[a], Xin Cheng[b], Xingliang Zheng[b,c], and Shengming Ma[a,b]*

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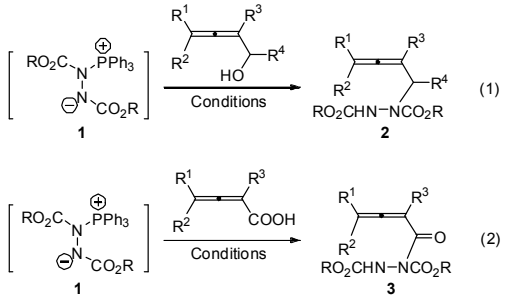
An efficient cascade reaction of 2,3-allenoic acids with dialkyl azodicarboxylate in the presence of PPh_3 affording 1,2-bis(alkoxycarbonyl)pyrazol-3-ones in moderate yields has been developed. The reaction may proceed via the intramolecular Michael addition of the expected 2,3-allenoyl hydrazines.

Pyrazol-3-ones and derivatives are important structural motif for agricultural chemistry and pharmaceutical development (Scheme 1),[1,2] which exhibits a wide range of biological properties as herbicides,[3] insecticides,[4] analgestic agents,[5] anti-inflammatory agents,[6] and therapeutic agents.[7] They have also been used as analytical reagents,[8] and dyes.[9] For these reasons, there are many reports dealing with the development of efficient synthetic methods for pyrazol-3-ones and derivatives.[10] The most widely used method is the cyclocondensation of β-keto alkanoates,[10] α-cyano esters,[11] β-enamino esters,[12] 2-alkenoates,[13] or 2-alkynoates[14] respectively with hydrazines. Although some progress has been achieved, some of these known protocols require harsh reaction conditions, multiple steps, or low efficiency, thus, development of mild, diverse, and efficient method is still highly desirable.



Scheme 1

"Huisgen zwitterion" formed from the reaction of azodicarboxylates with phosphines is an important intermediate in organic chemistry.[15] However, the chemistry of Huisgen zwitterions has received little attention until very recent years. A number of interesting reactions involving this zwitterion have been explored mainly by groups like Miller,[16] Lee,[17] Nair,[18] Hughes,[19] Wang,[20] and Shi[21] et al. Recently, we have found that the "Huisgen zwitterion" **1** can react with 2,3-allenols to give the 2,3-allenoyl hydrazines **2** (Scheme 2, eq 1), which further undergo palladium-catalyzed coupling-cyclization reaction with aryl iodides to afford *trans*-1,2-diazetidines[22] or 2,3-dihydro-1*H*-pyrazoles[23] depend on the substituent at the position 2 of 2,3-allenols. 2,3-Allenoic acid is an important class of allene derivatives.[24] The acidity and the polarized carbon-carbon double bond endow this kind of compounds a quite different reactivity as compared with 2.3-allenols. In conjunction with our continuous study on the chemistry of functionalized allenes, we envisioned the reaction of 2,3-allenoic acids with Huisgen zwitterion **1** may afford 2,3-allenoyl hydrazines **3**, which may have further synthetic utility for the synthesis of heterocyclic products (Scheme 2, eq 2).



Scheme 2

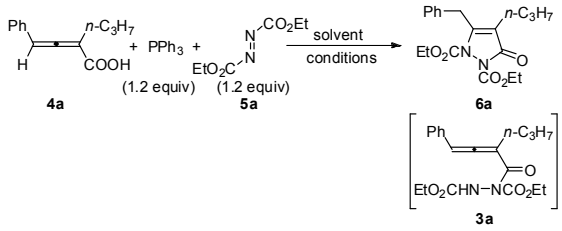
In our initial attempt, we investigated the reaction of 2,3-allenoic acid **4a** with diethyl azodicarboxylate **5a** in the presence of triphenylphosphine. The corresponding results under different conditions were summarized in Table 1. In THF, however, the reaction afforded an unexpected 1,2-dihydro-3*H*-pyrazol-3-one product **6a** in 63% NMR yield, which was fully characterized by $^1H/^{13}C$ NMR, IR, MS, and HRMS analysis (entry 1, Table 1). This must be formed from the intramolecular conjugate addition of the anticipated 2,3-allenoyl hydrazine product **3a**. It is obvious that the carbonyl group is triggering such a transformation. However, when the reaction was conducted at room temperature, the NMR yield of **6a** dropped to 50% (entry 2, Table 1). Adding more triphenylphosphine and diethyl azodicarboxylate failed to increase the yield (compare entry 3 with entry 1, Table 1), and shorting the reaction time to 0.5 h did not get better result (entry 4, Table 1). The reaction in dioxane also afforded the product **6a** in 63% NMR yield (entry 5, Table 1). More solvent effects were also examined in this reaction and the desired product **6a** can be obtained in moderate to excellent yields (entry 6-9, Table 1). Finally, the reaction condition (1.2 equiv. of PPh_3 and dialkyl azodicarboxylate, CH_2Cl_2, -10 ^oC 1 h then rt 1 h) was defined as the optimal set of conditions for this cyclization reaction (entry 9, Table 1).

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Table 1. Reaction of 2,3-Allenoic Acid **4a** with DEAD in the Presence of PPh_3 under Different Reaction Conditions



entry	solvent	conditions	NMR yield of **6a** (%)[b]
1	THF	-10 °C for 1 h then rt for 1h	63
2	THF	rt, 1 h	50
3[a]	THF	-10 °C for 1 h then rt for 1h	60
4	THF	-10 °C for 0.5 h then rt for 1 h	61
5	dioxane	-10 °C for 1 h then rt for 1h	63
6	Et_2O	-10 °C for 1 h then rt for 1h	42
7	DME	-10 °C for 1 h then rt for 1h	65
8	DCE	-10 °C for 1 h then rt for 1h	81
9	**CH_2Cl_2**	**-10 °C for 1 h then rt for 1h**	**84**
10	toluene	-10 °C for 1 h then rt for 1h	25[c]
11	Benzene	-10 °C for 1 h then rt for 1h	0[d]
12	*n*-Hexane	-10 °C for 1 h then rt for 1h	0[d]

[a] 1.5 equiv. each of PPh_3 and DEAD were added. [b] Determined with 1,3,5-trimethylbenzene as the internal standard. [c] Determined with CH_2Br_2 as the internal standard. [d] No expected products was detected.

Some typical results for the reaction of differently substituted 2,3-allenoic acids with PPh_3 and dialkyl azodicarboxylate under the optimized condition are listed in Table 2. As can be seen from Table 2, R^2 can be an alkyl group or an allyl group, while R^1 can be an aryl group, hydrogen, or an alkyl group. R^3 can be ethyl, isopropyl, *tert*-butyl, or phenyl group.

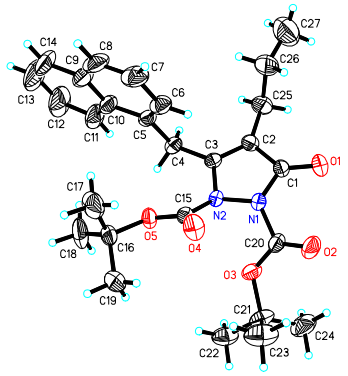
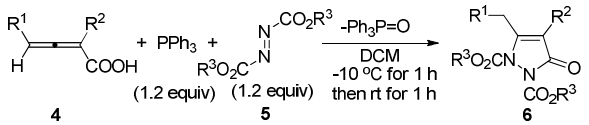


Figure 2. ORTEP representation of **6j**.

The 1,2-dihydro-3*H*-pyrazol-3-one structure of product **6** was further unambiguously established by the single crystal X-ray diffraction study of **6j** (Fig. 2).[25]

A plausible mechanism for the formation of **6** is shown in Scheme 3. The Huisgen zwitterion **1** formed from triphenylphosphine and dialkyl azodicarboxylate **5** reacts with

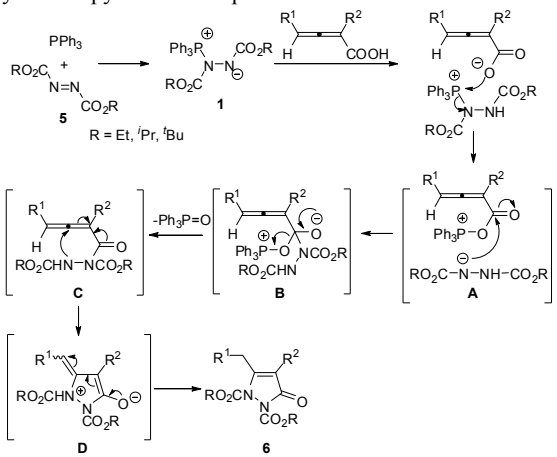
Table 2. Reaction of Dialkyl Azodicarboxylates and Various 2,4-Disubstituted or 2-Monosunstituted 2,3-Allenoic Acids in the Presence of PPh_3[a]



entry	substrate **4**		**5**	isolated yield of **6** (%)
	R^1	R^2	R^3	
1	Ph	*n*-Pr (**4a**)	Et (**5a**)	73 (**6a**)
2	Ph	*n*-Pr (**4a**)	*i*-Pr (**5b**)	84 (**6b**)
3	Ph	*n*-Pr (**4a**)	*t*-Bu (**5c**)	71 (**6c**)
4	Ph	*n*-Bu (**4b**)	Et (**5a**)	67 (**6d**)
5	Ph	*n*-Bu (**4b**)	*i*-Pr (**5b**)	65 (**6e**)
6	Ph	Me (**4c**)	Et (**5a**)	61 (61)[b] (**6f**)
7	Ph	Me (**4c**)	*i*-Pr (**5b**)	61 (**6g**)
8	1-naphthyl	*n*-Pr (**4d**)	Et (**5a**)	86 (**6h**)
9	1-naphthyl	*n*-Pr (**4d**)	*i*-Pr (**5b**)	86 (**6i**)
10	1-naphthyl	*n*-Pr (**4d**)	*t*-Bu (**5c**)	74 (**6j**)
11	1-naphthyl	Me (**4e**)	Et (**5a**)	64 (**6k**)
12	1-naphthyl	Me (**4e**)	Ph (**5d**)	66 (**6l**)
13	*p*-BrC_6H_4	*n*-Pr (**4f**)	Et (**5a**)	65 (**6m**)
14	*p*-BrC_6H_4	*n*-Pr (**4f**)	*t*-Bu (**5c**)	56 (**6n**)
15	*p*-BrC_6H_4	Allyl (**4g**)	Et (**5a**)	55 (**6o**)
16	H	Allyl (**4h**)	Et (**5a**)	58 (**6p**)
17	H	Bn (**4i**)	Et (**5a**)	54 (**6q**)
18	H	Me (**4j**)	Et (**5a**)	39 (**6r**)
19	Me	Me (**4k**)	Et (**5a**)	50 (**6s**)
20	C_3H_7	Allyl (**4l**)	Et (**5a**)	65 (**6t**)

[a] 1.2 equiv. of PPh_3 and dialkyl azodicarboxylate, CH_2Cl_2, -10 ℃ 1 h then rt 1 h. [b] Reaction conducted on one gram scale.

2,3-allenoic acids **4** to form oxyphosphonium salt **A**. substitution of **A** with hydrazine ion as nucleophilic agents gives intermediate **B**. **B** loses a molecule of triphenylphosphine oxide to afford 1,2-allenoyl hydrazine **C**,[19] which is so reactive that it instantly undergoes an intramolecularMichael addition to afford intermediate **D**. Subsequent protonransfer furnishes the final 1,2-dihydro-3*H*-pyrazol-3-one products **6**.



Scheme 3. Proposed Mechanism for the reaction

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In conclusion, we have developed a convenient cascade reaction of readily available 2,3-allenoic acids[26] with dialkyl azodicarboxlate in the presence of PPh_3 to afford 1,2-dihydro-3*H*-pyrazol-3-ones in moderate yields. A possible mechanism for this cascade reaction is proposed. Further studies on the scope and synthetic applications of this reaction are being conducted in our laboratory.

Acknowledgements

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Notes and references

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† Electronic Supplementary Information (ESI) available: Experimental procedures and characterization data of new compounds, copies of NMR spectra, and X-ray crystallography data. CCDC 715833. For EI and crystallographic data in CIF or other electronic format see DOI: 10.1039/b000000x/

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25 The crystal data of compound **6j:** $C_{27}H_{34}N_2O_5$; MW = 466.57, triclinic, monoclinic, space group *P*2(1)/n, Mo Kα, final R indices [$I > 2\sigma$(I)], R1 = 0.0650, *w*R2 = 0.1841, a = 14.4117 (13) Å, b = 11.1267 (10) Å, c = 16.3730 (15) Å, α = 90°, β = 93.689(2)°, γ = 90°, V = 2620.0 (4) Å^3, T = 293(2) *K*, Z = 4, reflections collected/unique: 13483/4874 (R_{int} = 0.0797), number of observations [$> 2\sigma(I)$] 4874, parameters 315. Supplementary crystallographic data have been deposited at the Cambridge Crystallographic Data Center. CCDC: 715833.

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